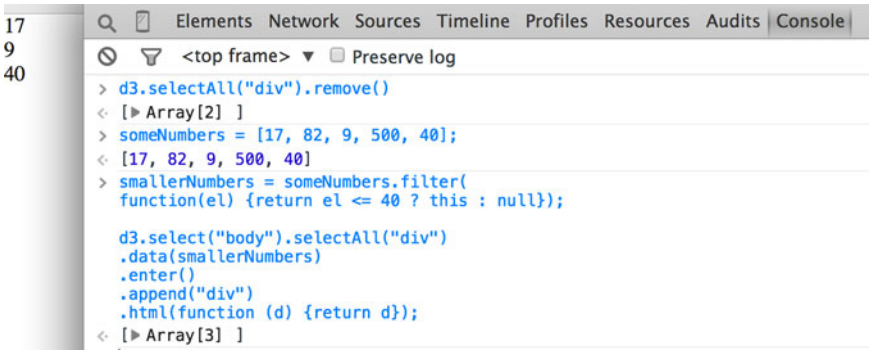


Figure 1.27 Running JavaScript in the console allows you to test your code. Here you've created a new array called `smallerNumbers` that consists of only three values, which you can then use as your data in a selection to update and create new elements.

anonymous function called when setting the `.style()`, `.attr()`, `.property()`, `.html()`, or other function of a selection can provide you with the data bound to that selection. As you explore examples, you'll see this function deployed again and again:

```
.style("background", function(d,i) {return d})
.attr("cx", function(d,i) {return i})
.html(function(d,i) {return d})
```

In every case, the first variable (typically represented with the letter *d*, but you can declare it as whatever you want) contains the data value bound to that element, and the second variable returns the array position (known as an index, hence the variable name *i*) of the value bound to that element. This may seem a bit strange, but you'll get used to it as you see it used in a variety of ways in the upcoming chapters.

JavaScript has many other array functions, and you can do much more than we covered here, but that's the subject of several other books. It's time to look at the kinds of data you'll work with.

1.4 Data standards

Standardization of methods of displaying data has been fed by and feeds into standardization of methods of formatting that data. Data can be formatted in a variety of manners for a variety of purposes, but it tends to fall into a few recognizable classes: tabular data, nested data, network data, geographic data, raw data, and objects.

1.4.1 Tabular data

Tabular data appears in columns and rows typically found in a spreadsheet or a table in a database. Although you invariably end up creating arrays of objects in D3, it's often more efficient and easier to pull in data in tabular format. Tabular data is delimited with a particular character, and that delimiter determines its format. You can have Comma-Separated Values (CSV), where the delimiter is a comma, or tab-delimited values, or a semicolon or a pipe symbol acting as the delimiter. For instance, you may